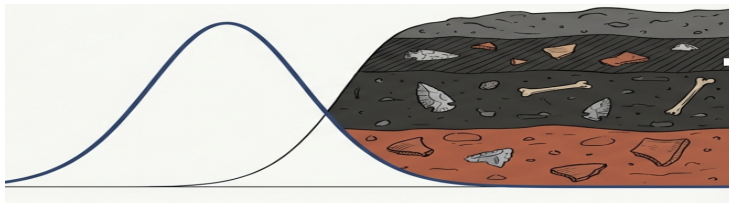


Bayesian Modeling of Dating Methods: From Physical Measurement to Historical and Geological Scenarios

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Plan

Introduction : the Bayesian principle

Individual age estimation

Event model : a robust way to combine measurements

Chronological model

A new approach : isotonic regression

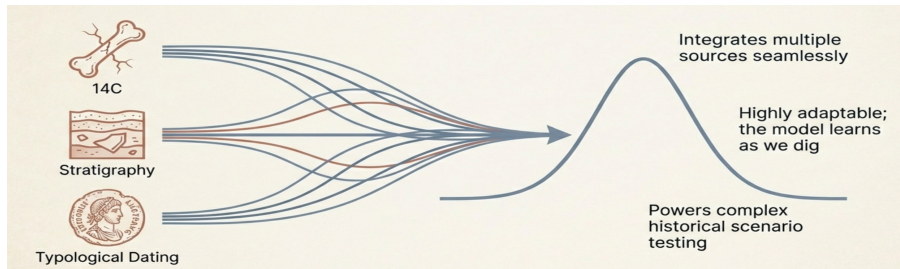
Post-processing of the Bayesian chronological model

reference

Why Bayesian Statistics in Archaeology ?

Objective : To explore how integrating diverse sources of data improves archaeological inference...

- ▶ Incorporate prior knowledge and new data.
- ▶ Sequentially update the estimates as new data become available.



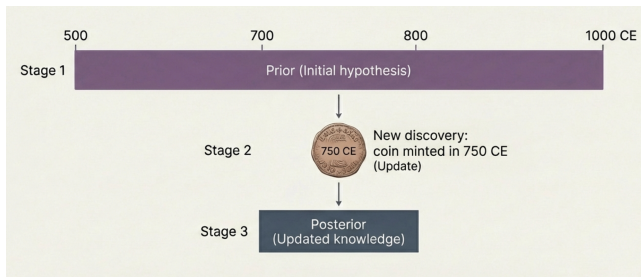
Principle of Knowledge Update

Key idea : An initial hypothesis (prior) is updated with new data.

Situation :

- ▶ A site is assumed to date from 500 to 1000 CE (high uncertainty).
- ▶ A coin struck in 750 CE is found.
- ▶ Update the probability distribution : the 700-800 CE range is favored.

Graphical representation :



Why Use Bayes in Archaeology ?

Advantages :

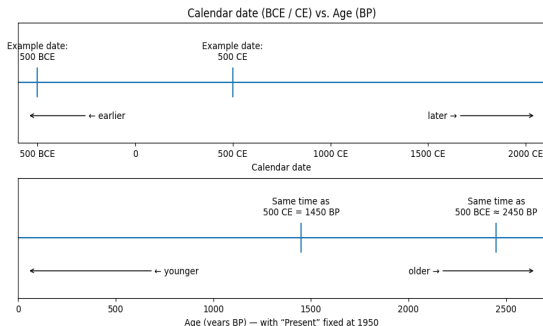
- ▶ Integration of multiple sources of information (stratigraphy, radiocarbon dating, artifacts).
- ▶ Adaptability to new discoveries.
- ▶ Incorporation of expert knowledge into the model.

Applications :

- ▶ Calibration of radiocarbon dating
- ▶ Determination of the most probable historical scenarios, taking into account uncertainties.
- ▶ Estimation of human migrations.
- ▶ Reconstruction of urban dynamics.

Age or Date ?

- ▶ Date : a point on the calendar scale (BCE/CE)
 - ▶ 500 CE = 500 ap J.C.
 - ▶ 500 BCE = 500 av. J.C.
- ▶ Age : the time elapsed before a reference year (Present = 1950 in radiocarbon dating,)
 - ▶ 500 BCE = 1450 BP (since $1950 - 500 = 1450$)
 - ▶ 500 CE = 2450 BP (since $1950 + 500 = 2450$)



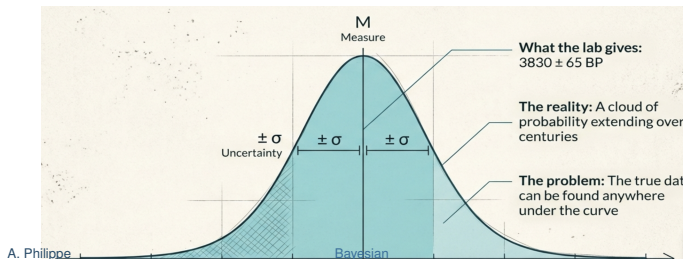
The physical measurements

- ▶ Each dating method provides a measurement with an associated uncertainty. $M \pm \sigma$
 - ▶ a ^{14}C age,
 - ▶ a paleodose measurement in TL/OSL,
 - ▶ a paleodose measurement in ESR
 - ▶ Uranium-thorium measurement

A measurement is not a single point but a probability distribution.

Interpretation of the measurement : $M \pm \sigma$

- ▶ M is a central value and σ is the dispersion around M
- ↪ Gaussian bell curve / Gaussian probability density function



$M \pm \sigma$ or $M \pm 2\sigma \rightsquigarrow$ same Gaussian bell Curve

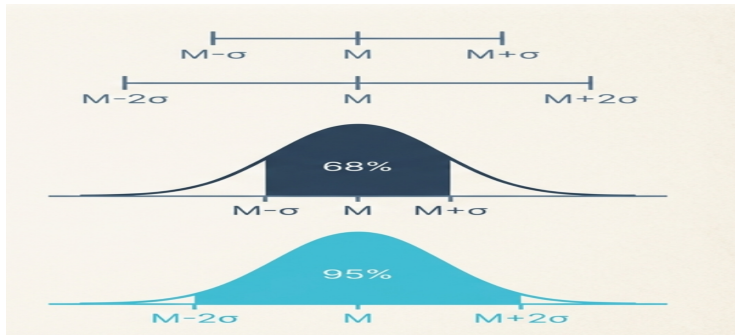
► The measurement $M = 2450$ and $\sigma = 30$

► $M \pm 1\sigma \rightarrow [2420, 2480]$

68% interval : $M \pm 1\sigma$

► $M \pm 2\sigma \rightarrow [2390, 2510]$

95% interval : $M \pm 2\sigma$

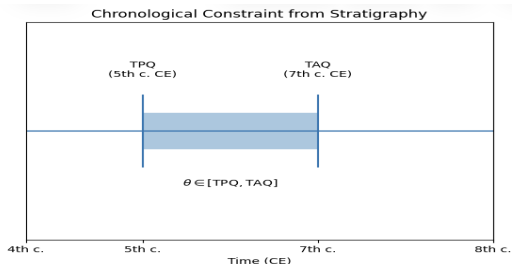


Archaeological information

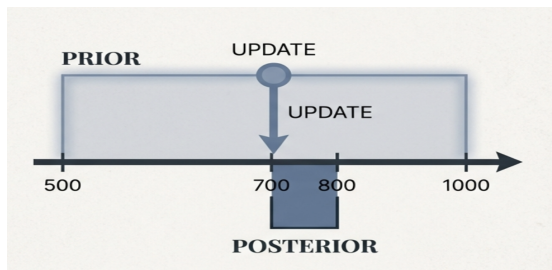
After the archaeological excavations, prior information is available on the dates.

Archaeological information : TPQ & TAQ .

- *Terminus Post Quem/ Terminus Ante Quem* for an event or artifact
 1. Terminus Post Quem : the TPQ indicates that the event or artifact must have occurred or was produced **after** this date,
 2. Terminus Ante Quem : the TAQ indicates that the event or artifact must have happened or been created **before** this date,

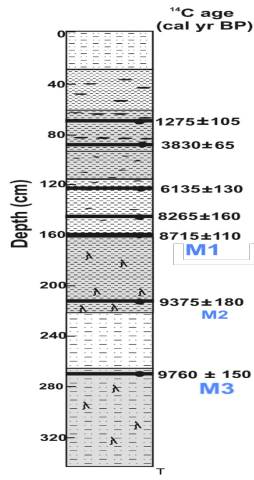


- ▶ Prior Knowledge : A site is assumed to date between 500 and 1000 CE.
- ▶ New evidence : a ceramic shard reveals specific traces of an 8th-century style $\rightsquigarrow$ TPQ & TAQ

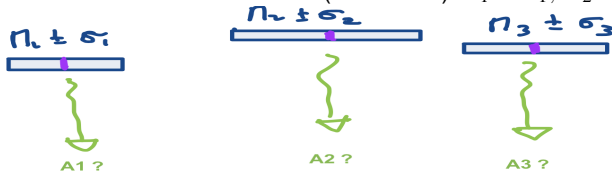


Archaeological information : Stratigraphy in sediment core

- **Stratigraphic information** induces an ordering of the dates.
lower layers are older than upper layers, assuming no major disturbance has occurred.

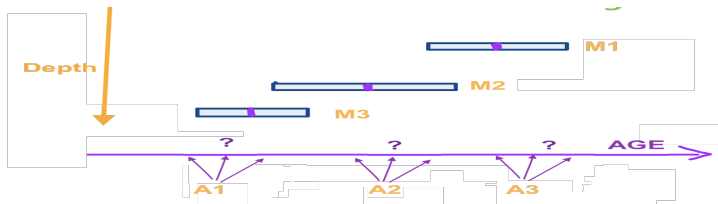


- Observations : N measurements (here $N=3$) $M_1 \pm \sigma_1, M_2 \pm \sigma_2, \dots, M_N \pm \sigma_N$



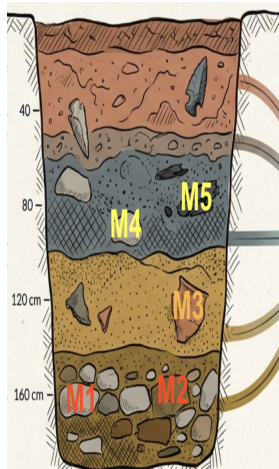
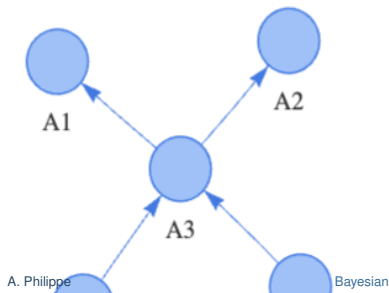
- $A_1, \dots, A_N$ is the unknown ages.
- Prior information on $A_1, \dots, A_N$:

$$A_1 \leq A_2 \leq A_3 \leq \dots \leq A_N$$



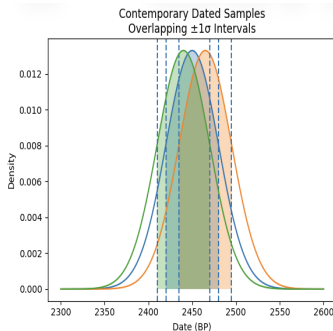
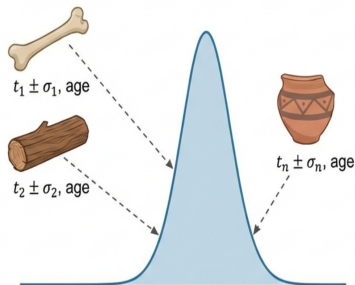
Archaeological information : General stratigraphic relationships

- ▶ Five measurements obtained from three layers.
- ▶ No chronological relationship between M_1 and M_2 .
- ▶ No chronological relationship between M_4 and M_5
- ▶ $\{Age_4, Age_5\} < \{Age_3\} < \{Age_1, Age_2\}$



Archaeological information : contemporaneity constraints

- Dated samples are contemporaneous : The analyzed samples have dates that come from the same historical or geological period.

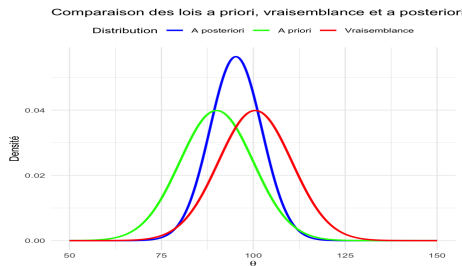
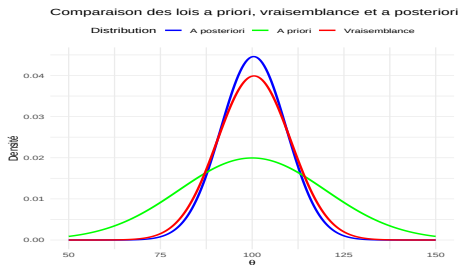


Fundamentals of Bayesian Statistics

Bayes Formula

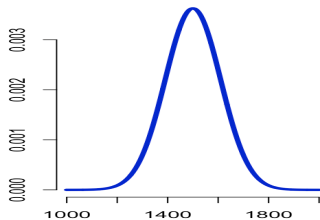
1. Observations $M \pm \sigma$, $\rightsquigarrow$ **Gaussian curve for the measurement**
2. prior information : $\rightsquigarrow$ **T Probability distribution on the date / age**
3. The posterior distribution of the date/ age

$$p^{posterior}(date \mid Measurement) \propto p(measurement \mid date) p^{prior}(date)$$

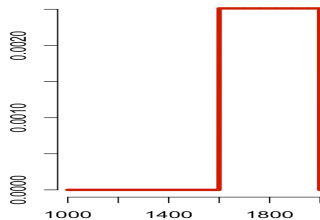


Example : TPQ & TAQ

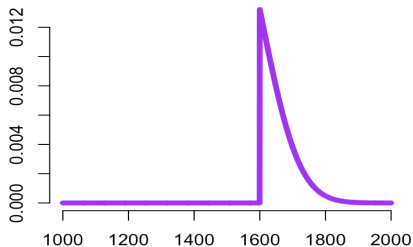
mesure 1500 +/- 110



a priori entre 1600 et 2000



loi a posteriori



Bayes Formula - 2 -

1. Observations $M_1 \pm \sigma_1, M_2 \pm \sigma_2, \dots, M_N \pm \sigma_N$

↪ Gaussian curve for each measurements $p(M_1, \dots, M_n | \theta_1, \dots, \theta_N)$

2. prior informations :

↪ Probability distribution on the dates/ ages $p^{prior}(\theta_1, \dots, \theta_N)$

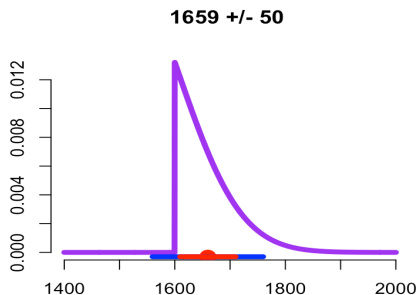
3. Bayes Formula : The posterior probability distribution on the dates/ ages

$$p^{posterior}(\theta_1, \dots, \theta_N | M_1, \dots, M_n) \propto p(M_1, \dots, M_n | \theta_1, \dots, \theta_N) \times p^{prior}(\theta_1, \dots, \theta_N)$$

Bayesian inference

From the posterior distribution, we can calculate

- ▶ the mean value
- ▶ the variance / Standard deviation



$A \pm \sigma$ is often an inadequate summary of the posterior distribution.

Bayesian inference - 2- Region of credibility

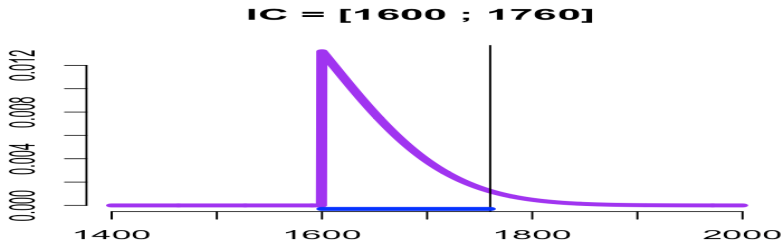
From the posterior distribution, we calculate

- Credible region in the Bayesian context : HPD region

$$P(\theta \in \text{HPD} | \text{measurements}) = 95\% (= 1 - \alpha)$$

with H as small as possible

- The regions with the highest density are selected

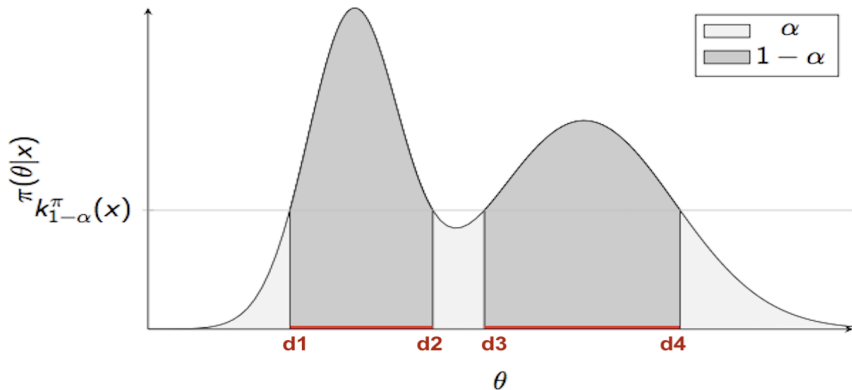


Statistical conclusion :

The posterior probability that the date lies in $[1600; 1760]$ is $1 - \alpha = 95\%$

Bayesian inference - 3-

Another example of posterior distribution with HPD region



Statistical conclusion :

The data belongs to the region

$$[d_1, d_2] \cup [d_3, d_4]$$

with probability $1 - \alpha = 95\%$

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Example 1 : Optically stimulated luminescence dating

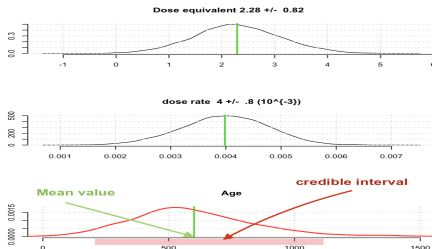
1. The measurements provided by the laboratory are :

- ▶ the equivalent dose $De \pm s$
- ▶ the dose rate $d \pm s_d$

2. Age calculation

$$A =: \frac{De}{d}$$

3. Prior information : the age is less than 1000 years.



Example 2 : Determination of radiocarbon age/date

The laboratory provides as measurement, the Age before calibration :

$$M = A^{BCAL} \pm \sigma$$

This Age before calibration is related to the « true » age/date of sample

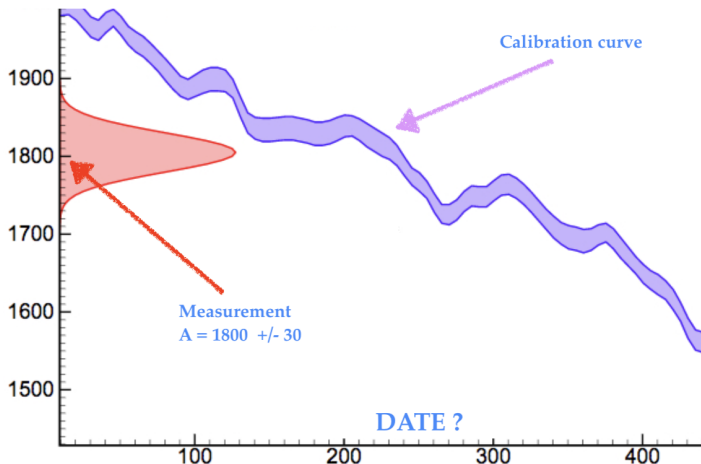
$$A^{BCAL} = g(\theta)$$

where

- ▶ θ is the calendar time or age to be calculated
- ▶ g is a calibration function which relates the measurement to *date*

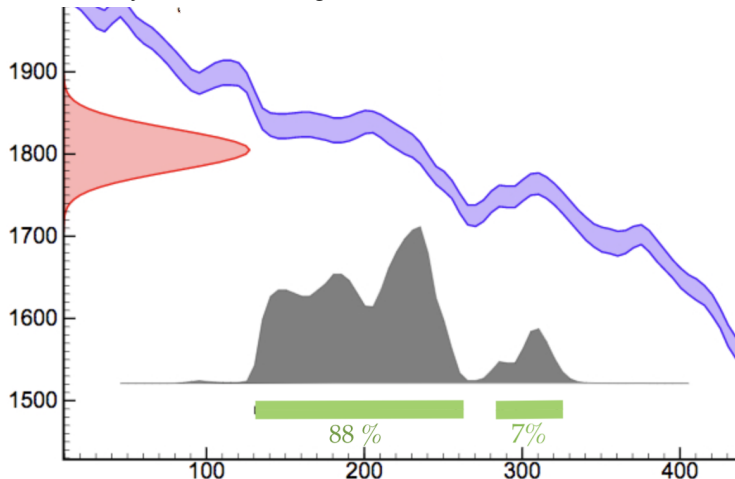
Calibration of radiocarbon

- ▶ The laboratory gives $A^{BCAL} \pm \sigma$ here 1800 ± 30
- ▶ Prior information : the date belongs to $[0, 500]$

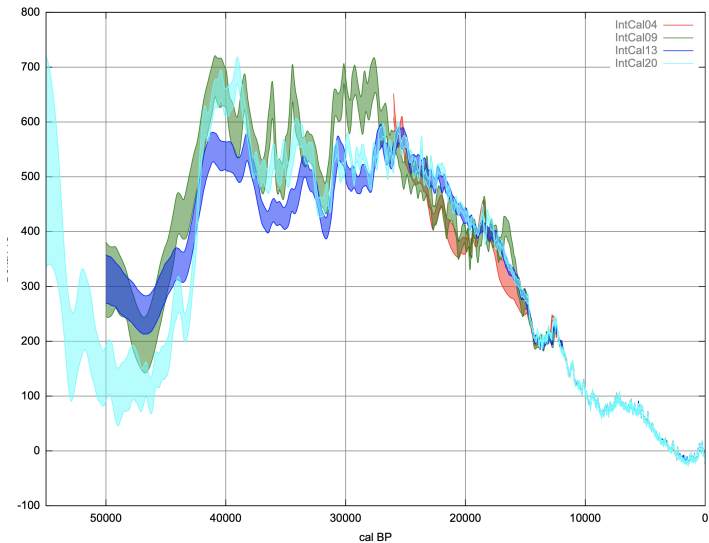


Calibration of radiocarbon

- ▶ The laboratory gives $A^{BCAL} \pm \sigma$
- ▶ Prior information : the date belongs to $[0, 500]$
- ▶ Result of the Bayesian modelling

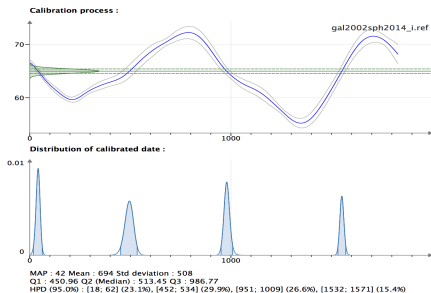


Calibration 14C : evolution of the curve

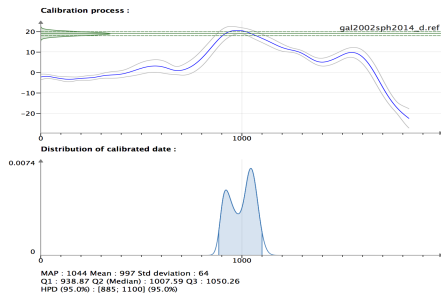


Exemple 3 : Archaeomagnetic calibration

inclinaison (AM)



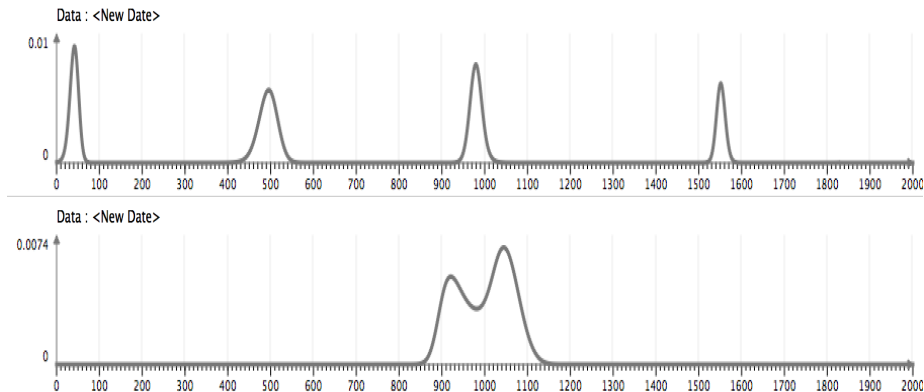
declinaison (AM)



Converting

- ▶ an inclination measurement (here $Incl = 65 \pm 1$) in calendar date via the calibration curve in France (Paris) over the last two millennia.
- ▶ an declination measurement (here $dec = 29 \pm 1$) in calendar date via the calibration curve in France (Paris) over the last two millennia.

Estimation of the date by two dating methods (Inclinaison / Declinaison)



How can we combine the information from both dating methods ?

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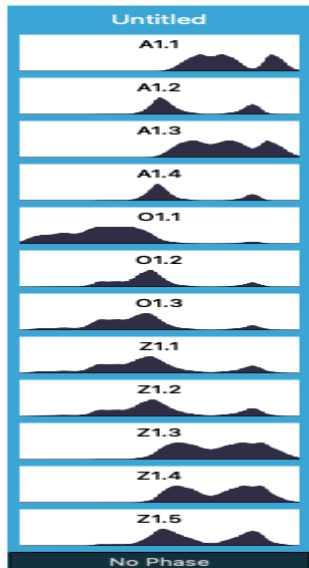
Definition of the target Event

Definition

- ▶ we choose a group of dated events that are related to the target event
- ≈ Characterize the date of a target event from the combination of the dates of contemporaneous dates.

The objective is to estimate the calendar date θ of the "target event"

The radiocarbon dating of the shroud of Turin



- ▶ 12 radiocarbon datings have been performed on a strip cut from the shroud of Turin
- ▶ The strip is divided into three samples sent to 3 AMS laboratories.
 - ▶ Arizona,
 - ▶ Oxford
 - ▶ Zurich
 - ▶ The shroud belongs to the historical period [0 , 2000].

How to determine the date of the shroud from the 14 individual calibrated dates ?

Medieval kiln of the potter's workshop in Lezoux

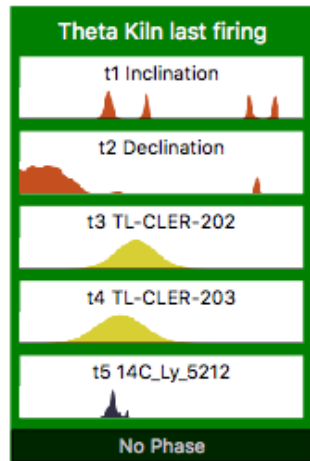


Dated artifacts :

- ▶ baked clays dated by
 - AM** > *Estimation of the last time the temperature exceeded a critical point*
 - TL** > *Estimation of the last firing*
- ▶ bones
 - 14C** > *Estimation of the death of the animal*

Assumptions :

- ▶ **All dated events are contemporaneous with the last firing of the kiln**
- ▶ This is any date between 0 and 2 000



Definition of the Event Model : central age/date model

Lanos & Anne Philippe (2017,2018+)

1. We want to estimate θ . the date of the target event or the central date
2. The target event is defined by
 - ▶ n measurements :
$$M_1 \pm \sigma_1, \dots, M_n \pm \sigma_n$$
 - ▶ Each measurement M_i is related to a unknown calendar date t_i
3. The prior information is
 - ▶ the date of the target event belongs to $\theta \in [T_b; T_e]$
 - ▶ all the dates $t_1, \dots, t_n$ are contemporaneous

The statistical model

The model is

$$M_i = g_i(t_i) \pm \sigma_i$$

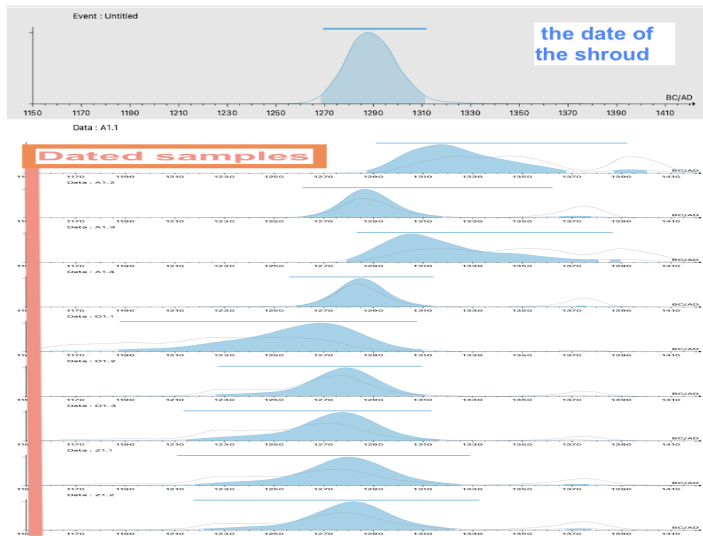
$$t_i = \theta \pm \delta_i$$

$$\theta \sim \text{Uniform}(T)$$

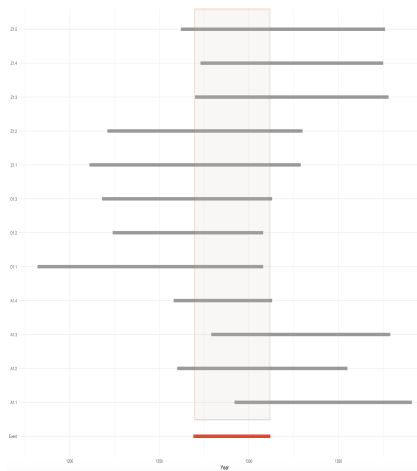
- ▶ σ_i represents the experimental and calibration error
- ▶ δ_i represents the difference between the date of artifacts t_i and the target event θ

All the dates $t_1, \dots, t_n$ are close to the central date θ but with possible errors corresponding to $\delta_1, \dots, \delta_n$

Radiocarbon dating of the shroud of Turin (1) : posterior distributions



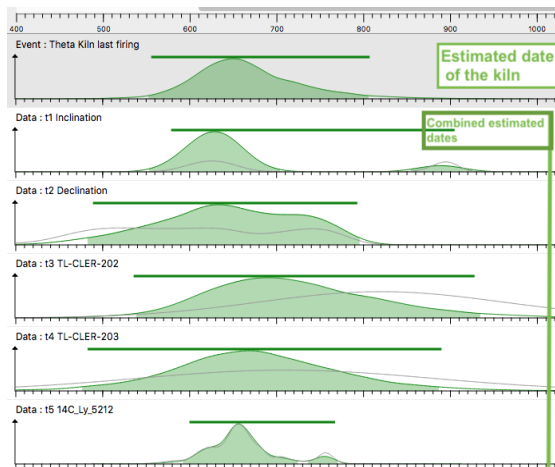
Radiocarbon dating of the shroud of Turin (2) : credible Interval 95%



From Scattered Evidence to Precision :

- ▶ Event Model : Combines individual densities into a single distribution
- ▶ Robust estimation : Reduces overall uncertainty
- ▶ Final event date : Greater concentration around the true date

Kiln of the potter's workshop in Lezoux



Test of the assumption of contemporaneity

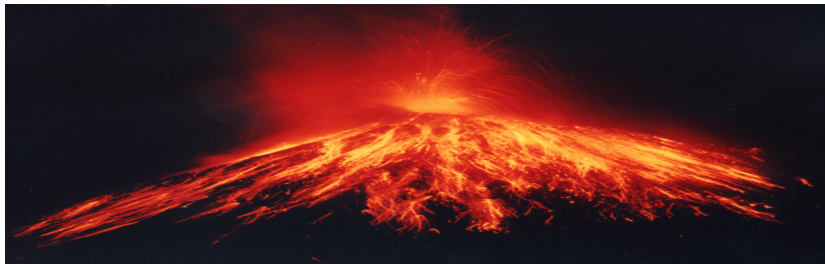
- ▶ To test the assumption of contemporaneity, we analyze the error between the target event θ and dated events t_i .
- ▶ Individual standard deviations σ_i :

$$\theta = t_i \pm \sigma_i$$

Analysis of σ_i distributions

- ▶ If all densities are concentrated on small values
 $\rightsquigarrow$ the assumption of contemporaneity is true
- ▶ Large deviations correspond to potential outliers and are automatically penalized in the estimation of the event date.

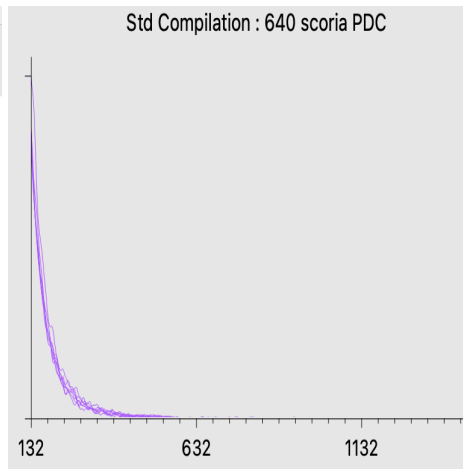
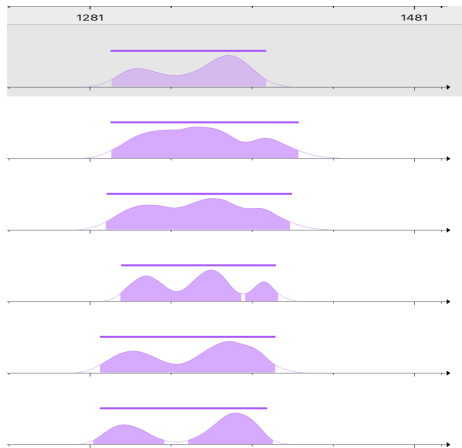
Volcanic eruptions



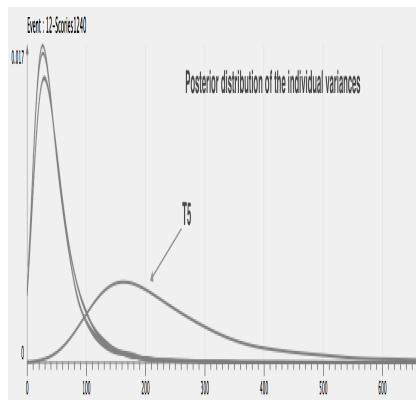
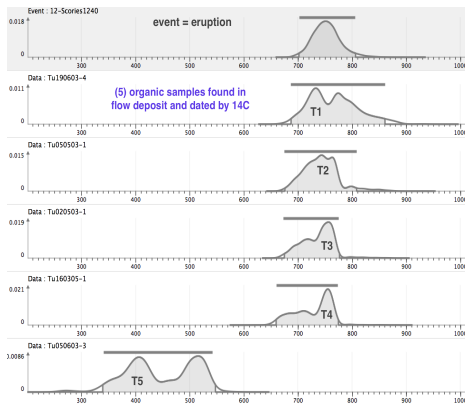
- ▶ **Target Event** : Eruptive period with flow deposits
- ▶ **Dated events** : organic samples found in a flow deposit are dated by ^{14}C .

A first pyroclastic flow

- ▶ Target event : eruption [θ]
- ▶ 5 organic samples found in flow deposit are dated by ^{14}C [$t_1, \dots, t_5$]



A pyroclastic flow containing an outlier.



- ▶ the posterior density of date of the target Event remains almost insensitive to the outlier.
- ▶ We do not have to choose specific tools for rejecting outlying data.

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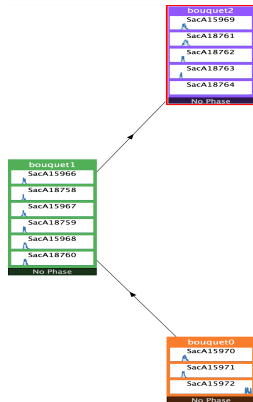
We consider Bayesian tools for constructing chronological scenarios.

Main idea of the model implemented in `Chronomodel`

1. we define target event as a group of contemporaneous dated events.
2. We construct a chronology (= collection of dates) of target events taking into account temporal relationship between the dates of target events

Alternative : model implemented in `Oxcal`

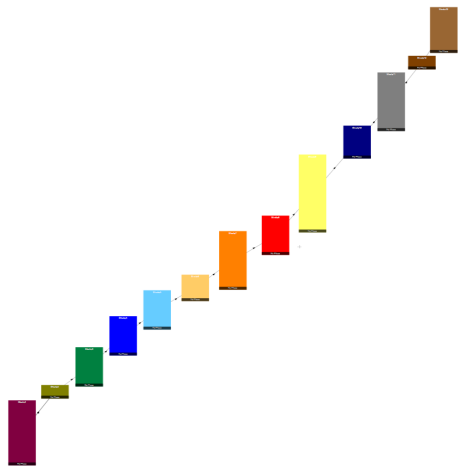
- We construct a chronology of dates of target events



Volcanic eruptions



- ▶ **Target Event** : Eruptive period with flow deposits
- ▶ **Dated artefacts** : organic samples found in a flow deposit are dated by ^{14}C .
- ▶ **Prior information** Stratigraphic constraint on deposits



Restrictions

- ▶ Each event contains at least one measurement.
- ▶ Each measurement is associated to one (and only one) target event.

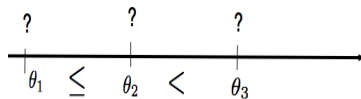
Chronologies of K target events

- We want to estimate $\theta_1, \dots, \theta_K$ the calendar dates of target events.

Prior information on the dates of the target event

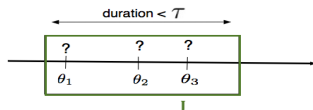
1. The stratigraphic constraints.

$\rightsquigarrow$ a partial uniform order on
 $(\theta_1, \dots, \theta_K) := \vartheta \subset T^K$



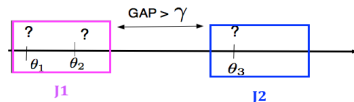
2. Duration information :

$\max_{j \in J} \theta_j - \min_{j \in J} \theta_j \leq \tau$ where τ is known



3. Hiatus information :

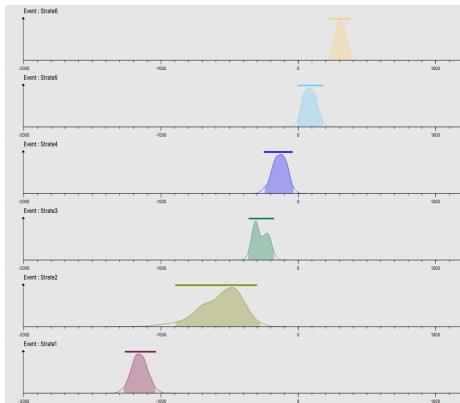
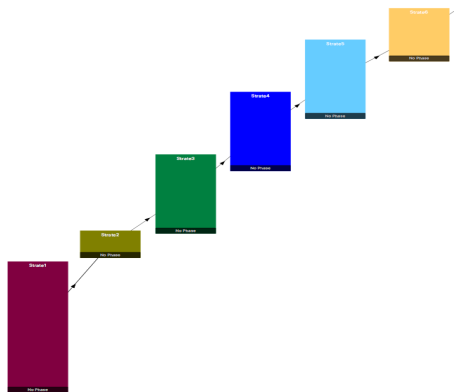
J_1, J_2 two groups, $\min_{j \in J_2} \theta_j - \max_{j \in J_1} \theta_j \geq \gamma$
 where γ is known



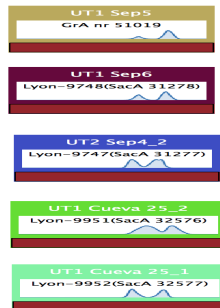
Chronology of Volcanic eruptions

6 pyroclastic flows from volcano dated by ^{14}C $\rightsquigarrow$ 6 ordered target events

$$S = \{\vartheta : \theta_1 \leq \dots \leq \theta_6\}$$



Maya city with information on occupation time



Prior information on the archaeological phase :

The occupation duration is less than 50 years..

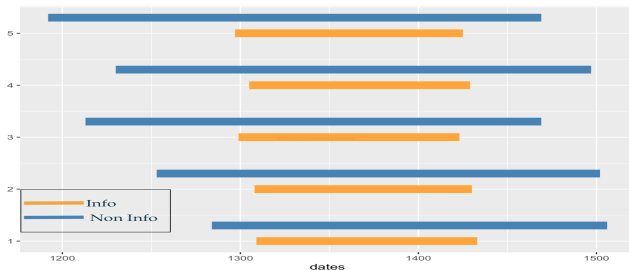
Effect of information on precision

without information

event	interval	length
θ_1	1284 - 1506	222
θ_2	1253- 1502	249
θ_3	1213- 1469	256
θ_4	1230- 1497	267
θ_5	1192- 1469	277

Information on the duration

event	interval	length
θ_1	1309 - 1433	124
θ_2	1308-1430	122
θ_3	1299 - 1423	124
θ_4	1305 - 1429	124
θ_5	1297 - 1425	128



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Event model : a robust way to combine measurements

Chronological model

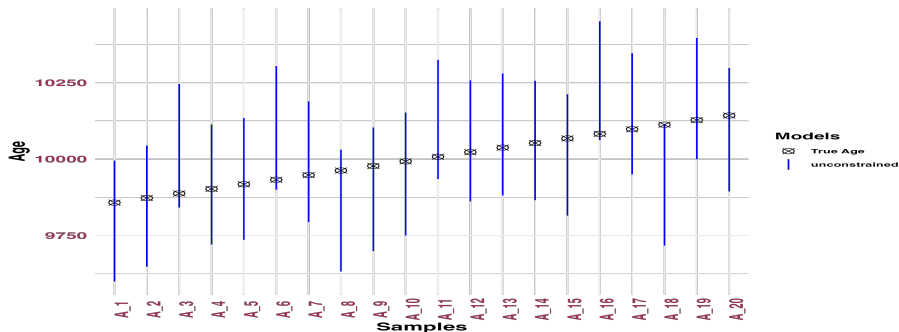
A new approach : isotonic regression

Post-processing of the Bayesian chronological model

reference

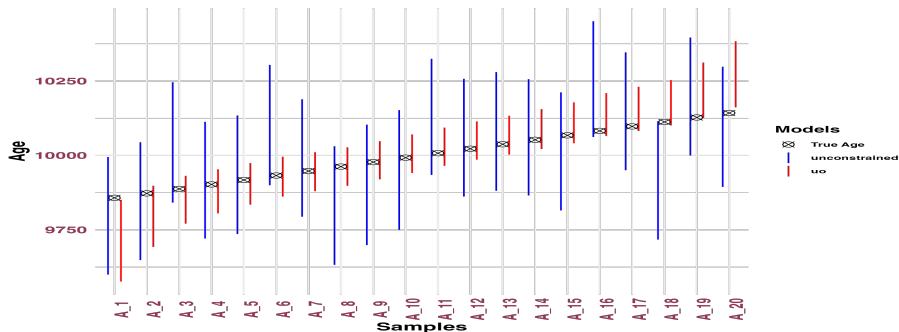
High-frequency measurements

1. We have N measurements $M_1, \dots, M_N$ over a short time interval $[T_1, T_2]$.
2. We estimate the corresponding ages $A_1, \dots, A_N$ using a Bayesian model, incorporating the prior information that all ages lie within the interval $[T_1, T_2]$.



3. Additional information : stratigraphic ordering in a sediment core, leading to the constraint

$$A_1 \leq \dots \leq A_N.$$



In high-frequency data, a bias may occur, affecting both the smallest and the largest ages.

Alternative Approach : isotonic regression

Instead of forcing stratigraphic rules at the beginning, we first let the data speak for itself, and then we "map" it onto the stratigraphy using a method called Isotonic Regression

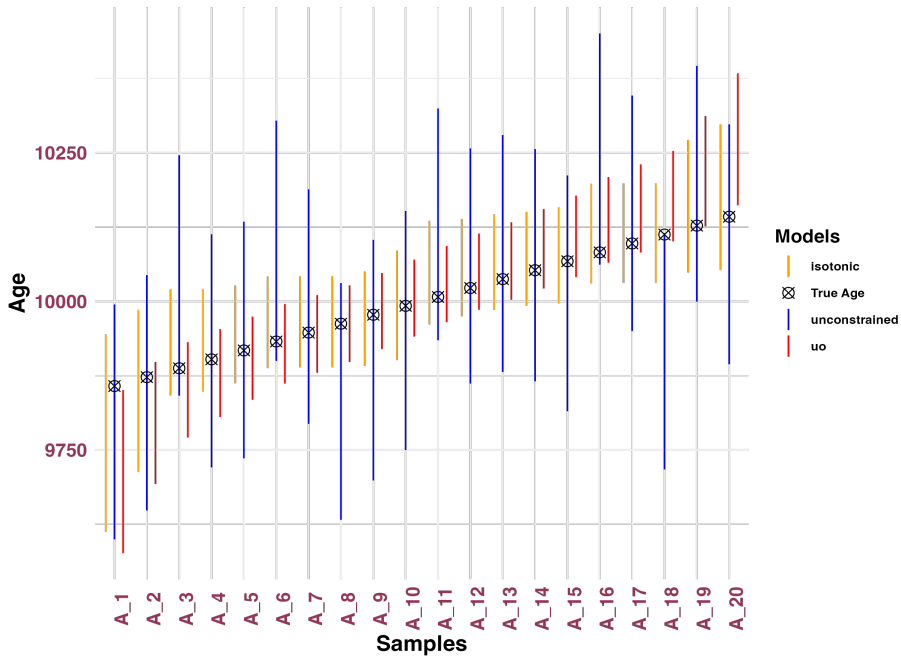
The steps :

1. We first estimate the corresponding ages $A_1, \dots, A_N$ using a Bayesian model, incorporating the prior information that all ages lie within the interval $[T_1, T_2]$.
2. We compute the projection of the posterior distribution onto the stratigraphic constraint set :

$$A_1 \leq \dots \leq A_N.$$

In other words, we select the probability distribution closest to the estimated one that satisfies the ordering constraints.

3. This yields the least informative distribution satisfying the constraints.



Plan

Introduction : the Bayesian principle

Individual age estimation

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Statistical analysis of the chronology

Examples

1. Characterisation of a group of dates [begin / end /duration/ period]
2. Testing the presence of hiatus between two dates or two groups of dates.
3. Construction of a tempo plot to evaluate the repartition in time

The R package ArcheoPhase : contains Statistical Tools for analysis the chronological modelling

Age-depth model

Additional information : the depth of the dated event.

1. We estimate the relation between the dates t and the depth h

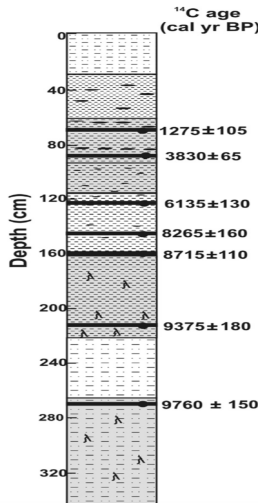
$$f(t) = h \quad \text{age-depth curve}$$

2. We estimate f taking into

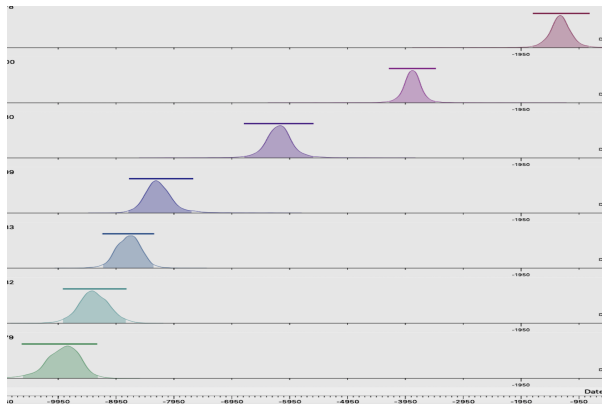
- ▶ all the posterior information on the sequence of dates estimated by the Bayesian chronological model
- ▶ Non parametric regression method is applied on the output of the MCMC algorithm.

3. From the estimated curve, we predict the date as function of the depth.

Stratigraphy and radiocarbon ages of the lake sediments.



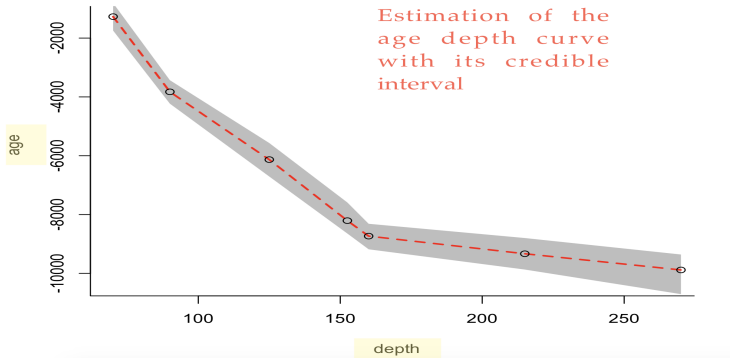
- Information on the date of events : temporal order coming from the stratigraphy.
- Result of the Bayesian modeling



Estimation of Age -depth curve

To estimate this curve we use

1. estimated ages by the chronological model
2. the depth of the dated samples

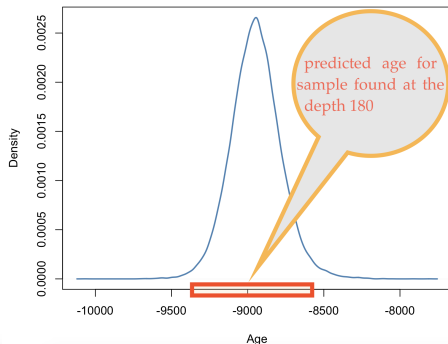
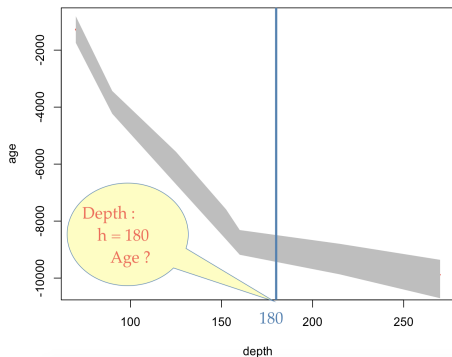


Forecasting

We want to predict the age of a sample that would be collected at a selected depth h

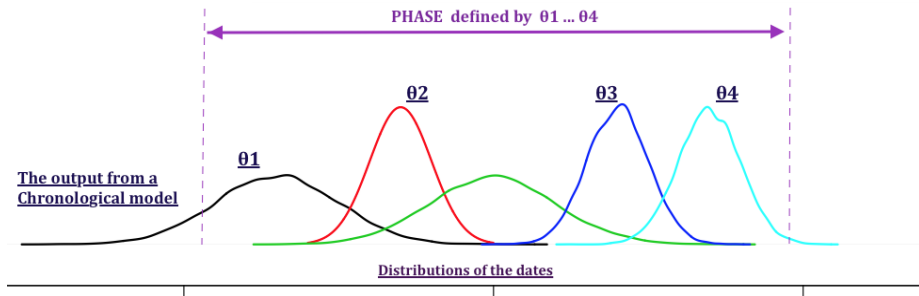
Method

1. The estimation is based on the estimated age-depth curve
2. We take into account the uncertainty on the estimated curve



Phases : definition

A phase is a group of dates defined on the basis of objective criteria such as archaeological, geological or environmental criteria.



The collection of dates is estimated from a chronological model.
[Chronomodel / Oxcal ...]

$$\text{Phase} = \{\theta_j, j \in J \subset \{1, \dots, K\}\}$$

Estimation of the phase

$$\text{Phase}_1 = \{\theta_j, j \in J \subset \{1, \dots, K\}\}.$$

1. posterior distribution of the minimum

$$\alpha = \min_{j \in J} \theta_j$$

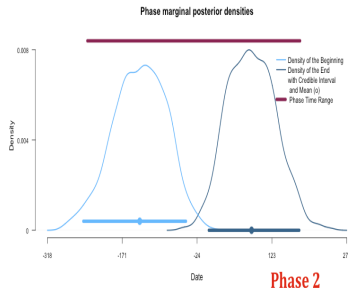
$\rightsquigarrow$ Estimation of the beginning

2. posterior distribution of maximum

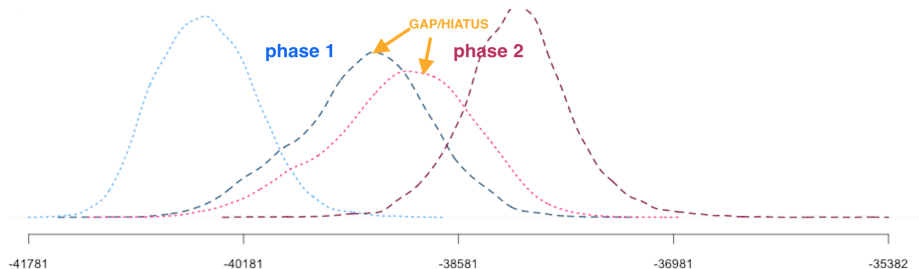
$$\beta = \max_{j \in J} \theta_j \rightsquigarrow \text{Estimation of the end}$$

3. **Phase time range** The shortest interval that covers all the dates θ_j included in the phase at level 95%
i.e. the shortest interval $[a, b] \subset T$ such that

$$P(\text{for all } j \theta_j \in [a, b] | M_1, \dots, M_n) = P(a \leq \alpha \leq \beta \leq b | M_1, \dots, M_n) = 95\%$$



Hiatus



Detection of a hiatus between two phases $\theta_j, j \in J_1$ and $\theta_j, j \in J_2$

1. $\beta_1 = \max_{j \in J_1} \theta_j$ and $\alpha_2 = \min_{j \in J_2} \theta_j$
2. Can we find $[c, d]$ such that

$$P(\beta_1 < c < d < \alpha_2 | M_1, \dots, M_n) = 95\%$$

The chronology of Canímar Abajo in Cuba

(Rocksandic *et al.* 2015 Philippe & Vibet (2018) RadioCarbon.

The site has evidence for two episodes of burial activity separated by a shell midden layer.

- ▶ 12 AMS radiocarbon dates (human bones collagen and a charcoal) obtained from burial contexts
- ▶ 7 from the Older Cemetery (OC),
- ▶ 5 from the Younger Cemetery (YC)

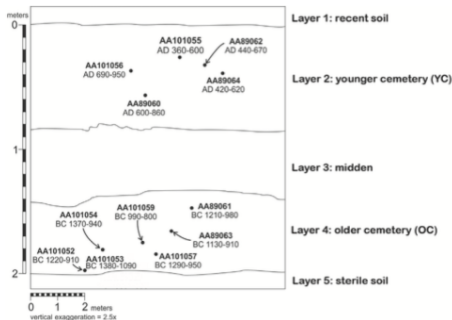
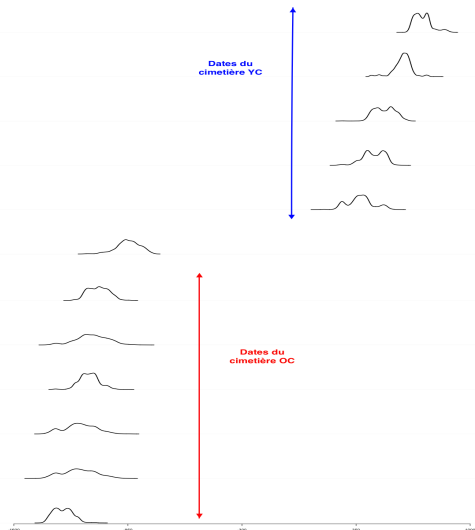


Figure 2 Stratigraphic profile indicating relative positions of samples for AMS ^{14}C dating

The aim : Bayesian model based on these 12 AMS radiocarbon dates in order to draw conclusions about

- ▶ the time of both mortuary activities
- ▶ the hiatus between them

The chronology

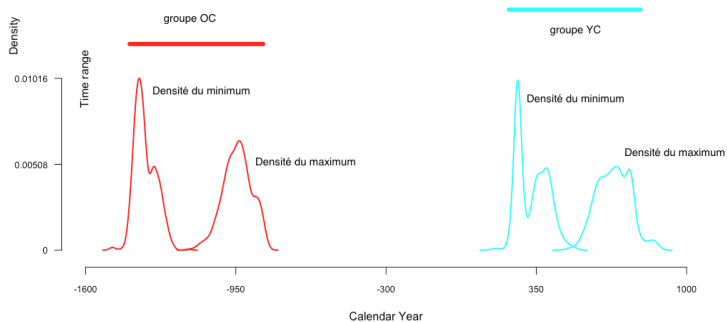


From the estimation of the sequence of dates $t_1, \dots, t_{12}$ (using Bayesian model) we estimate

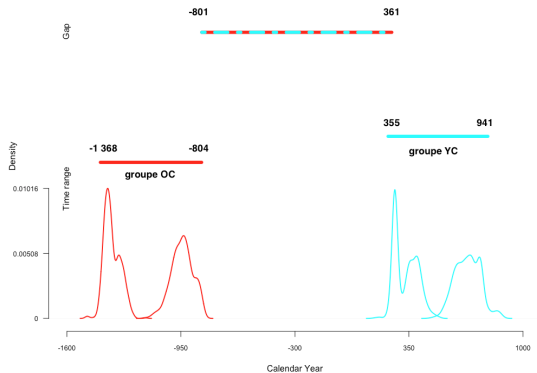
- ▶ the beginning and the end of the Older Cemetery
- ▶ the beginning and the end of the Younger Cemetery
- ▶ the gap between these two periods

Estimation of the dates t

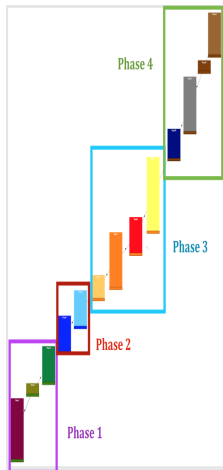
Chronology of the activities in the site of Canimar Abajo.



Estimation of the hiatus



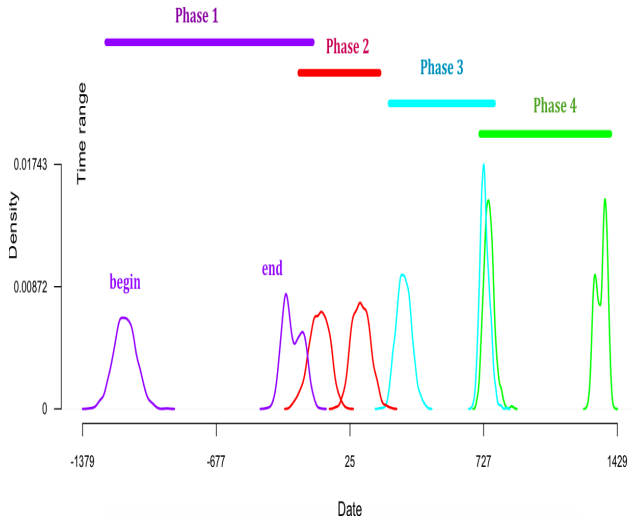
Application to Volcanic eruptions [cont]



$$P_1 = \{\theta_1, \theta_2, \theta_3\}, \dots$$

$$P_4 = \{\theta_{10}, \theta_{11}, \theta_{12}, \theta_{13}\}$$

Phases marginal posterior densities

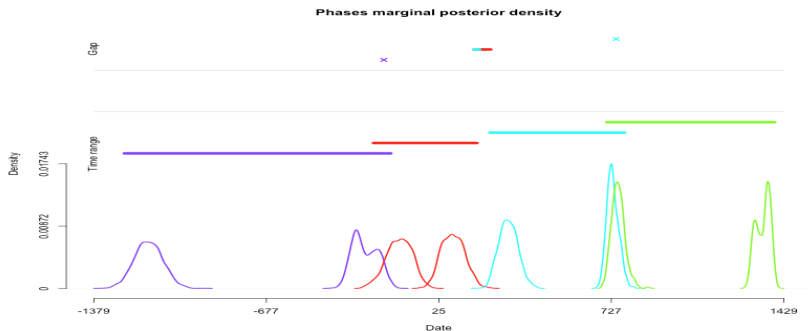


Application.

Detection of hiatus :

- ▶ A hiatus is detected between Phases 2 & 3.
Estimation of the interval [170, 235]
- ▶ there is no gap between 1 & 2 and 3 & 4

To summarise



Study of rhythms.

(see Dye 2016 and Philippe & Vibet 2017)

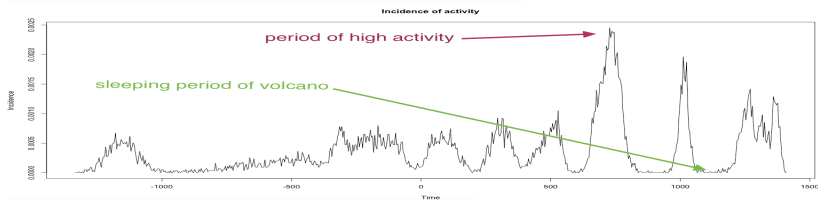
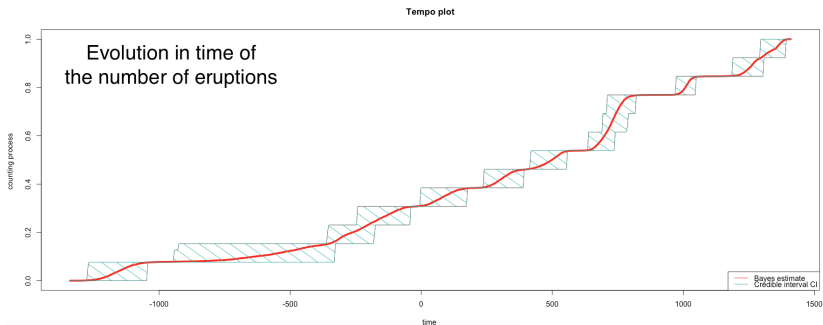
A statistical graphic designed for the study of rhythms.

- ▶ The tempo plot measures change over time :
- ▶ For each date t , we estimate the number of events $N(t)$ which occurs before the date t , we have

$$N(t) = \sum_{i=1}^n \mathbb{I}_{]-\infty, t]}(\tau_i)$$

- ▶ Interpretation : the slope of the plot directly reflects the pace of change :
 - ▶ a period of rapid change yields a steep slope
 - ▶ a period of slow change yields a gentle slope.
 - ▶ When there is no change, the plot is horizontal.

Application : Evaluation of the activity of volcano



Analysis of East Texas Woodland period¹

Aim : Exploring the development and geographical continuity of the peoples in East Texas

Material :

Target events : evidence of human activity

Dated events : death of nutshells captured along with the deposit of human activity

Measurements : radiocarbon dating

Three areas

Sulphur River : 18 measurements

Sabine River : 10 measurements

Neches River : 35 measurements

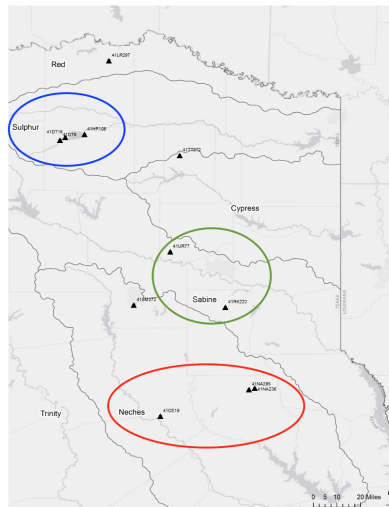
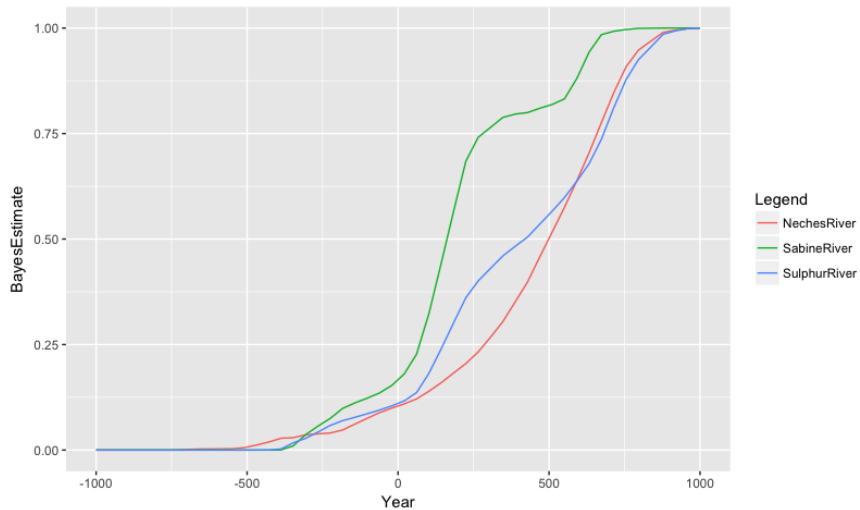


Figure 1 Map of east Texas river basins and the 11 Woodland period sites with 4 or more ¹⁴C dates.

1. Selden (2012). Modeling regional radiocarbon trends : a case study from the East Texas Woodland period. Radiocarbon Vol 54.

Comparisons of tempo plot



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Softwares

1. **Oxcal** provides radiocarbon calibration and analysis of archaeological and environmental chronological information.

Bronk Ramsey, C. (1995). Radiocarbon calibration and analysis of stratigraphy The OxCal program. Radiocarbon, 37(2), 425-430.

2. **Chronomodel**

Lanos, A. Philippe (2017) Hierarchical Bayesian modeling for combining Dates in archaeological context. Journal de la SFdS, Vol. 158 (2) pp 72-88.

Lanos and Philippe (2019) Event date model a robust Bayesian tool for chronology building. Communications for Statistical Applications and Methods

R software

1. **ArchaeoPhases** Post-Processing of the Markov Chain Simulated by 'ChronoModel', 'Oxcal' or 'BCal'

Philippe, Anne, and Vibet Marie-Anne 2020. Analysis of Archaeological Phases using the CRAN Package ArchaeoPhases. Journal of Statistical Software 93 (Code Snippet 1) : 1-25.

2. **BayLum** **BayLumPlus**. Chronological Bayesian Models Integrating Optically Stimulated Luminescence and Radiocarbon Age Dating

Imène Bouafia, Anne Philippe, Guillaume Guérin. A Bayesian Framework for Parameter Estimation under Partial Order Constraints using Isotonic Regression. 2025..

B. Combes, A. Philippe. Bayesian analysis of individual and systematic multiplicative errors for estimating ages with stratigraphic constraints in optically stimulated luminescence dating. Quaternary Geochronology 39, 2017.

A. Philippe, G. Guerin S. Kreutzer, BayLum an R package for Bayesian Analysis of OSL Ages & Chronological Modelling (LED2017)

3. **Luminescence** Comprehensive Luminescence Dating Data Analysis